



A COMMUNITY OF CHOICE

CITY ENGINEER'S OFFICE
2101 O'Neil Avenue
Cheyenne, Wyoming 82001

MEMO

TO: Honorable Mayor Patrick J. Collins, City of Cheyenne
Members of the City Council, City of Cheyenne

FROM: Wesley D. Bay, P.E., City Drainage Engineer

DATE: May 6, 2025

SUBJECT: Professional Service Agreement between the City of Cheyenne and GLM Design Group, for a feasibility study, benefit-cost analysis, and preliminary design of the Cheyenne Dry Creek Union Pacific Railroad Project.

Please find the attached Professional Service Agreement between the City and GLM Design Group for engineering services for a feasibility study, benefit-cost analysis, and preliminary design of the Cheyenne Dry Creek Union Pacific Railroad Project.

The project is located along Dry Creek, both upstream and downstream of the Union Pacific Railroad (UPRR). The Dry Creek Drainage Basin is approximately 14.6 square miles and extends from Warren Airforce Base to I-80 east of Cheyenne. Development in basin has increased impervious area and associated stormwater runoff. In a significant storm event, water impounds against the 25' high UPRR track embankment east of Pershing Blvd and is released under the tracks through a historic 6' by 8' by 75' masonry culvert. The impoundment of water against the UPRR embankment has led to more stringent drainage requirements in the basin than other basins in the city.

The project will include a feasibility analysis related to the boring of several culverts under the UPRR embankment and increased detention both west and east of the UPRR tracks. Several conveyance and detention scenarios will be investigated and evaluated. The Dry Creek Master Plan Update (DMP) lists this as the top priority project for the Dry Creek Drainage basin.

Engineering Staff evaluates fee proposals for engineering services based on time and material estimates for the specific project. Those costs are readily tracked and verified during design operations and the City only pays for the actual time expended. We believe this agreement will provide the required services to complete analysis and preliminary design portion of the project.

Funding for this project will come from the 2023 1% Specific Purpose Tax Fund (5th Penny) and a Federal Emergency Management Agency (FEMA) Building Resilient Infrastructure and Communities (BRIC) grant. The total professional services will be a cost not-to-exceed \$246,974.80. We respectfully request your approval of this professional services agreement.

City Contract No. _____

PROFESSIONAL SERVICES AGREEMENT
for
Cheyenne Dry Creek Union Pacific Railroad Crossing Project

This Professional Services Agreement (“Agreement”) is made and entered into by and between the City of Cheyenne, located at 2101 O’Neil Avenue, Cheyenne, Wyoming 82001 (“City”), and GLM Design Group, LLC, P.O. Box 20505, Cheyenne, Wyoming 82003 (“Consultant”). This Agreement shall become effective upon the date of the last signature affixed hereto.

RECITALS

In consideration of the mutual covenants and obligations herein expressed, it is agreed by and between the parties as follows:

1. SCOPE OF WORK

The Consultant agrees to provide services as described for the above-mentioned project, in accordance with the Scope of Work attached hereto as Exhibit A, incorporated by reference and expressly made a part hereof.

2. TIME OF PERFORMANCE

This Agreement shall commence on the date the last required signature is attached hereto and shall continue in full force and effect until the scope of work is completed in accordance with Exhibit A, unless this Agreement is earlier terminated pursuant to its provisions.

3. COMPENSATION

In consideration of the services to be performed pursuant to this Agreement, the Consultant will bill the City, and the City agrees to pay Consultant a not to exceed amount of Two-Hundred Forty-Six Thousand Nine-Hundred Seventy-Four Dollars and Eighty Cents (\$246,974.80) as stated in Exhibit A.

4. CONSULTANT RESPONSIBILITY

In providing services under this Agreement, the Consultant shall perform in a manner consistent with that degree of care and skill ordinarily exercised by members of the same profession currently practicing under similar circumstances. Upon notice to the Consultant, the Consultant will, without additional compensation, correct those services not meeting such a standard.

5. COMPLIANCE WITH LAWS

The Parties agree that they will perform their obligations as provided in this Agreement in accordance with all applicable laws and ordinances.

6. DRUG-FREE WORKPLACE

In compliance with the Drug-Free Workplace Act of November 1988, the City of Cheyenne has established an Alcohol and Controlled Substance Policy that pertains to alcohol and drug usage by City Employees. All architects, engineers, and other consultants under contract with City of Cheyenne, and their employees and sub-consultants, are required to comply with the provisions of the City's Alcohol and Controlled Substance Policy for drug and/or alcohol usage on City property or other sites occupied by the Consultant while performing the duties and responsibilities of the Agreement. It is the responsibility of the Consultant to familiarize themselves with the requirements of this policy and to inform all their employees and sub-consultants of the requirements and ensure their compliance. If the Consultant, their employees or sub-consultants are found in violation of this policy, the Agreement may be terminated.

7. INDEMNIFICATION/HOLD HARMLESS

The Consultant agrees to indemnify, hold harmless, and defend the City from and against any and all liabilities, claims, penalties, forfeitures, and suits, and the cost and expenses incident thereto, including reasonable attorney's fees, which may hereafter arise as a result the performance of the Consultant's duties, including death or bodily injury to any person, destruction or damage to any property, contamination of or adverse effects on the environment, or any violation of governmental laws, regulations, or orders to the extent caused by (1) the Consultant's breach of any term or provision of this Agreement; or (2) any negligent or wrongful act, error, or omission by the Consultant, or its employees or sub-consultants in the performance of this Agreement. The Consultant acknowledges that it may incur a financial obligation to the City pursuant to the terms of this paragraph.

8. INSURANCE REQUIREMENTS

The Consultant must provide proof of the following insurance coverages:

Commercial General Liability Insurance

For claims arising out of bodily injury, illness, or death, or from damage to or destruction of property of others, including loss or use thereof, with minimum limits of \$1,000,000 per occurrence, \$2,000,000 aggregate for the entire term of the Agreement.

Workers' Compensation

Workers' compensation coverage shall be in effect for the entire term of the Agreement, as required by Wyoming law, for all employees or agents providing services under this Agreement. Consultant shall provide the City with proof of workers' compensation or employer's liability insurance coverage.

Professional Liability Insurance

The Consultant shall provide proof of professional liability insurance or errors and omissions liability insurance in an amount not less than \$500,000 to protect the City from any and all claims arising from the Consultant's negligence in the performance of duties under this Agreement. The City prefers that this liability insurance coverage be provided pursuant to an "occurrence" policy.

If this coverage is provided pursuant to a "claims made" policy:

- a. Consultant shall, concurrently with the execution of this Agreement, provide the City with a certificate of insurance demonstrating that such coverage is or shall be in effect at the time the Consultant begins the provision of services under this Agreement; and
- b. In the event the Consultant's services extend into a future policy period, the Consultant shall, prior to the policy expiration date, provide the City with a new certificate of insurance demonstrating that such coverage is or shall be in effect during all periods of time that Consultant will provide services under this Agreement; and
- c. Consultant shall maintain said "claims made" coverage for a period of five (5) years following the last date that Consultant has provided services under this Agreement; and
- d. In the event the Consultant or the insurer terminates "claims made" coverage prior to the expiration of the periods provided in subparagraphs a., b., or c. of this paragraph, the Consultant shall provide to the City advance written notification of the termination of said coverage and shall provide the City with an endorsement for an extended reporting period ("tail coverage") which shall be in effect for a period of time not less than five (5) years following the last date that Consultant has provided services under this Agreement.

Additional Insurance Information

The Consultant shall name the City of Cheyenne as **an Additional Insured** by endorsement on its insurance policies and shall provide the City with a copy of the endorsements. This requirement does not apply to workers' compensation and professional liability insurance policies.

Consultant shall provide the City with certificates of insurance acknowledging the above-stated coverages prior to beginning any work under this Agreement.

It is understood and agreed that these policies are primary and not contributory. All policies required under this Agreement shall be in effect for the duration of the Agreement. It shall be an affirmative obligation upon Consultant to immediately notify in writing the city risk manager, city clerk, and city attorney of any fact, circumstance, or

occurrence that has resulted in or may result in the cancellation of, or substantive change to any insurance coverage required by this Agreement, and failure to do so shall be construed to be a breach of this Agreement.

If requested by the City, the Consultant shall provide the City with copies of insurance policies and/or policy endorsements listing the City of Cheyenne as an additional insured. The City's failure to request or review such policies, endorsements, or certificates shall not affect the City's rights or Consultant's obligation hereunder.

Any insurance company providing coverage under this Agreement shall have a minimum A. M. Best rating of A- (excellent).

9. MINORITY AND DISADVANTAGED BUSINESS ENTERPRISES (49 C.F.R. Part 26)
All parties to this Agreement assure that no person will be excluded from participation in, denied the benefits of, or otherwise discriminated against, in connection with the award and performance of this Agreement on the grounds of age, race, color, disability, national origin, or sex.
10. MONTHLY REPORT
With every monthly billing, Consultant shall provide the City Representative with a written statement of the status of the work with respect to the Scope of Work, time sheets, and work schedule. Failure to provide the required monthly report will delay processing of any payment request until the report is submitted.
11. INDEPENDENT CONSULTANT
The Consultant shall function as an independent consultant for the purposes of this Agreement. The Consultant shall assume sole responsibility for any debts or liabilities that may be incurred by the Consultant in fulfilling the terms of this Agreement. Nothing in this Agreement shall be interpreted as authorizing the Consultant or its agents or employees to act as an agent or representative of or on behalf of the City or to incur any obligation of any kind on the behalf of the City.
12. TAXES
The Consultant agrees to pay all valid taxes, excises, license fees, permit fees, bills, debts, and obligations incurred by and in connection with its operations under this Agreement.
13. DEFAULT
Each and every term and condition in this Agreement shall be deemed to be a material element of the Agreement. In the event either party should fail or refuse to perform according to the terms of this Agreement, such party may be declared in default.

14. REMEDIES

In the event a party has been declared in default, such defaulting party shall be allowed a period of fifteen (15) days within which to cure the default. In the event the default remains uncorrected, the non-defaulting party declaring default may elect to:

- a. Terminate the Agreement and seek damages, which damages shall not exceed the contract amount; or
- b. Treat the Agreement as continuing and require specific performance.

15. TERM AND TERMINATION

The term of this Agreement shall be from date of execution until the provisions of the scope of work as presented in Exhibit A have been met. The parties may by mutual written agreement renew or extend this Agreement.

The City may, without cause, and upon thirty days' written notice to the Consultant, terminate this Agreement in whole or in part at any time, for the City's convenience. Upon receipt of such notice, the Consultant shall:

- a. Discontinue all services affected; and
- b. Deliver to the City within five (5) days all data, drawings, specifications, reports, estimates, summaries, and such other information and materials as may have been accumulated by the Consultant in performing this Agreement, whether completed or in process.
- c. In the event of termination for convenience, the City will pay the Consultant for accepted work done and expenses incurred to the date of termination. Such acceptance shall not be unreasonably withheld.

16. ADDITIONAL REMEDIES

In the event the Consultant fails to strictly perform in accordance with this Agreement, the City may elect to correct the deficiencies and charge the Consultant. In the event of default of any of the conditions by either party which shall require the party not in default to commence legal or equitable action against the defaulting party each party shall bear its own costs and expenses, including without limitation, attorney's fees.

17. GOVERNING LAW, JURISDICTION AND VENUE

The construction, interpretation, and enforcement of this Agreement shall be governed by the laws of the State of Wyoming. The courts of the State of Wyoming shall have jurisdiction over any action arising out of this Agreement and over the parties, and the venue shall be the First Judicial District, Laramie County, Wyoming.

18. GOVERNMENTAL IMMUNITY

The City of Cheyenne, along with its officials and employees, does not waive its governmental immunity by entering into this Agreement, except to the extent necessary for the parties to pursue a contract action to clarify or enforce the written terms of the Agreement. Furthermore, the City specifically retains all immunities and defenses

available to it as a sovereign or governmental entity pursuant to Wyo. Stat. § 1-39-101, *et seq.*, and all other relevant state and federal law. Designations of venue, choice of law, enforcement actions, and similar provisions should not be constructed as a waiver of governmental immunity. The parties agree that any ambiguity in this Agreement shall not be strictly construed, either against or for either party, except that any ambiguity as to governmental or sovereign immunity shall be construed in favor of governmental immunity.

19. MONITORING ACTIVITIES

The City shall have the right to monitor all activities related to this Agreement that are performed by the Consultant or its sub-consultants. This shall include, but not be limited to, the right to make site inspections at any time and with reasonable notice; to bring experts and consultants on site to examine or evaluate completed work or work in progress; to examine the books, ledgers, documents, papers, and records pertinent to this Agreement; and to observe personnel in every phase of performance of the related work.

20. OWNERSHIP OF DOCUMENTS AND WORK PRODUCT

All deliverables by the Consultant, all work product of the Consultant, all notes, calculations, memoranda, or any other documentation generated by the Consultant pursuant to the scope of this Agreement, shall be the property of the City of Cheyenne and shall be provided to the City of Cheyenne immediately upon request.

21. NON-DISCRIMINATION

The Consultant shall comply with the Civil Rights Act of 1964, the Wyoming Fair Employment Practices Act (Wyo. Stat. § 27-9-105, *et seq.*), the Americans With Disabilities Act (ADA), as amended, 42 U.S.C. § 12101, *et seq.*, and the Age Discrimination Act of 1975 and any properly promulgated rules and regulations thereto, and shall not discriminate against any individual on the grounds of age, sex, creed, color, race, religion, national origin, ancestry, pregnancy or qualifying disability in connection with the performance under this Agreement.

22. FORCE MAJEURE.

The performance of the Agreement by either party shall be subject to force majeure including, but not limited to, acts of God, fire, flood, natural disaster, war or threat of war, acts or threats of terrorism, civil disorder, unauthorized strikes, governmental regulation or advisory, recognized health threats as determined by the World Health Organization, the Centers for Disease Control, or local government authority or health agencies (including, but not limited to, the health threats of COVID-19, H1N1, or similar infectious diseases), curtailment of transportation facilities, or other similar occurrence beyond the control of the parties, where any of those factors, circumstances, situations, or conditions or similar ones prevent, dissuade, or unreasonably delay the performance required by this Agreement. The Agreement may be cancelled by either party, without

liability, damages, fees, or penalty, and any unused deposits or amounts paid shall be refunded, for any one or more of the above reasons, by written notice to the other party.

23. PROFESSIONAL REGISTRATION

The Consultant shall endorse, as required by law, plans and reports prepared under this Agreement and shall affix thereto his or her seal of professional registration, showing that he or she is licensed to practice in the State of Wyoming, if necessary.

24. SUPPLEMENTAL GENERAL CONDITIONS

In consideration of the mutual covenants and obligations herein expressed, it is further agreed by and between the parties as follows:

Standard of Care. The standard of care for all professional engineering and related services performed or furnished by Engineer under this Agreement will be the care and skill ordinarily used by members of the subject profession practicing under similar circumstances at the same time and in the same locality. The Engineer makes no warranties, express or implied, under this Agreement or otherwise, in connection with any services performed or furnished by Engineer.

Technical Accuracy. Owner shall not be responsible for discovering deficiencies in the technical accuracy of Engineer's services. Engineer shall correct deficiencies in technical accuracy without additional compensation unless such corrective action is directly attributable to deficiencies in Owner-furnished information.

Conflicts within the Agreement. If there is a conflict within this Agreement, the order of priority of the documents which make up this Agreement, from highest to lowest, will be:

- (a) Agreement.
- (b) Any supplementary conditions to the General Conditions.
- (c) General Conditions.
- (d) Schedule A – SCOPE OF SERVICES.
- (e) Other schedules to this Agreement.

Any changes in the Consultant's or Subconsultant's staff or fee structure will be presented in writing to the City for approval prior to initiating any changes or performing any work.

Overhead and Normal Business Expenses. The consultant shall include customary overhead costs and statutory benefits as part including, but not limited to, social security contributions, unemployment, excise and payroll taxes, workers' compensation, health and retirement benefits, bonuses, sick leave, vacation, and holiday pay, as applicable, within the billing rates furnished in Exhibit A. Additionally, the cost of general and administrative overhead which includes salaries and wages of employees engaged in

business operations not directly chargeable to projects, plus non-Project operating costs, including but not limited to, business taxes, legal, rent, utilities, office supplies, insurance, and other operating costs shall not be directly billed to project and assumed to be included within the billing rates furnished.

Design Process Documentation. To document the design process, the consultant will be required to provide a Decision and Design Log for the project duration. This log shall be furnished to the City of Cheyenne in conjunction with the project meetings. The purpose of the log is to document and track major decisions during the design and development process. The log shall generally follow the format outlined below in Table 1. Design and

Decision Log:

Table 1. Design and Decision Log						
No.	Date	Decision	Alternatives Considered	Reasons	Decision Made By	Additional Task Required
Sequenced log number.	Date the decision was made.	A one-sentence summary of the decision made.	A list of the other approaches considered.	A two to three sentence summary of why the decision was made. A link to additional documentation, if required and [Title].	Who made this decision?	If applicable, a list of additional tasks required by the design or follow-up required.

25. ENTIRETY OF AGREEMENT

This Agreement consisting of Nine (9) pages, and Exhibit A consisting of Four (4) pages, contain the entire understanding of the parties. There are no other terms or conditions, written or oral, concerning or controlling this matter.

[SIGNATURES OF FOLLOWING PAGE]

IN WITNESS WHEREOF, the Parties to this Agreement, through their duly authorized representatives, have executed this Agreement on the days and dates set out below and certify that they have read, understood, and knowingly and voluntarily agreed to the terms and conditions of this Agreement. This Agreement may be executed in multiple counterparts, each of which shall be deemed an original, but all of which together shall constitute one and the same instrument.

City of Cheyenne

(SEAL)

By: _____
Patrick Collins, Mayor

Attest:

Kylie Soden, City Clerk

Date: _____

Consultant, GLM Design Group, LLC

By: _____

Print Name: _____

Date: _____

PROFESSIONAL SERVICES AGREEMENT

Project Name: **Cheyenne Dry Creek Union Pacific Railroad Crossing (RFLP 1-25)**

Exhibit A – Scope of Work

Overview

Consultant services for the Cheyenne Dry Creek Union Pacific Railroad Crossing are the development of a feasibility study, benefit cost analysis, and preliminary design plans for additional stormwater detention in Kiwanis Park and additional conveyance under the Union Pacific Railroad (UPRR). Any innovative solutions that are cost effective and meet project goals will be incorporated during project scoping negotiations and included in the analyses/design. Work required for this project will include project management; field surveys, topographic data development, hydrologic/hydraulic analysis, and benefit cost analysis.

The project requires a professional design team that includes a Civil Engineer registered in the State of Wyoming with a minimum of 10 years of project design experience in one or more of the following disciplines: hydrologic/hydraulic analysis, and benefit cost analysis for major stormwater projects; with the ability to provide in-house or capacity to sub-consult for survey services and those disciplines required for the project.

Project Description

The project is located along Dry Creek, both upstream and downstream of the Union Pacific Railroad (UPRR). The Dry Creek Drainage Basin is approximately 14.6 square miles and extends from Warren Airforce Base to I-80 east of Cheyenne. Development in basin has increased impervious area and associated stormwater runoff. In a significant storm event, water impounds against the 25' high UPRR track embankment east of Pershing Blvd and is released under the tracks through a historic 6' by 8' by 75' masonry culvert. The impoundment of water against the UPRR embankment has led to more stringent drainage requirements in the basin than other basins in the city.

The project will include a feasibility analysis related to the boring of several culverts under the UPRR embankment and increased detention both west and east of the UPRR tracks. Several conveyance and detention scenarios to be investigated are presented later in this scope of work.

The Dry Creek Master Plan Update (DMP) and the East Park Greenway Underpass and Drainage Plan at the Union Pacific Railroad documents provide conceptual detention layouts and culvert configurations at the now named Kiwanis Park for one of the scenarios to be investigated. The DMP lists this as the top priority project for the Dry Creek Drainage basin.

PROFESSIONAL SERVICES AGREEMENT

Project Name: **Cheyenne Dry Creek Union Pacific Railroad Crossing (RFLP 1-25)**

Project Tasks and Deliverables

The expected project tasks and project deliverables are as follows:

Task 1: Feasibility Analysis

- Feasibility analyses will be performed and will include in-depth discussions of the advantages and disadvantages of each scenario. Ultimately, the Consultant will provide recommendations on the preferred alternative. The following alternatives should be considered, at a minimum:
 - Increase detention in Kiwanis Park to minimize flow increases through the proposed culvert configuration under the UPRR tracks. This alternative will minimize downstream impacts through detention and attenuation solely on the west side of the UPRR line.
 - Allow for higher flow through the proposed culverts under the UPRR tracks and mitigate downstream impacts east of the UPRR line with minimal increases to detention west of the UPRR line.
 - Increase detention in Kiwanis Park with increased flow through the proposed culverts under the UPRR tracks and mitigation of downstream impacts east of the UPRR line. Mitigation of flooding will be accomplished on both sides of the UPRR line.
 - Optionally, incorporate any other mitigative measures not specifically identified in this scope of work.

Task 2: Benefit Cost Analysis (BCA)

- Provide a Benefit Cost Analysis (BCA) for the preferred alternative including a comparison to the existing un-modified site. Benefits should include the following, at a minimum:
 - Benefits to the City and Laramie County.
 - The potential for expanded future development in the Dry Creek Basin.
 - Benefits to the UPRR and interstate commerce.

Task 3: Preliminary Design

- Produce a preliminary plan (approximately 30% plan) of the preferred alternative. The plans should include the following, at a minimum:
 - Plan and profile of channel realignment.
 - Plan and profiles of conveyance structures.
 - Plan and cross sections of proposed detention features and dams.
 - Preliminary quantities and estimation of project probable cost.

Implicate to the afore-mentioned tasks will be those steps necessary to complete the feasibility analyses, BCA, and preliminary plans. This will include, at a minimum, the evaluation and collection of topographical data; gathering of required geotechnical data; evaluation and supplementation of existing hydrologic and hydraulic models; preliminary design of the overall site, detention features, dams, and culverts under the UPRR tracks; preliminary construction cost estimation; and data gathering supporting benefit calculations.

PROFESSIONAL SERVICES AGREEMENT

Project Name: **Cheyenne Dry Creek Union Pacific Railroad Crossing (RFLP 1-25)**

Project Funding and Timing

The city has \$200k of funding available for the project. A portion of funds will be from a FEMA Building Resilient Infrastructure and Communities (BRIC) grant and the remaining portion will be from the 1% optional sales tax (5th Penny).

The grant period of performance expires December 29, 2025. The final feasibility analysis, recommendations, BCA, and preliminary design plans will be completed on or before this date.

General Conditions

The following items are included as part of the basic compensation for this project:

- Normal business expenses - phone, postage, etc.
- Cost of insurance.
- In-house computer time and service.
- Word processing, accounting, and man-hour records.
- Permits and license fees.
- Mileage.
- Travel fees, room and board, per diem.
- Printing costs for all standard review, bidding, and as-constructed plans and other correspondence and contract documents.
- Film and processing.

Additional items which will be addressed:

All subconsultant fees at cost, which may include:

- Geotechnical Investigation.
- Surveying.
- Utility investigation.

Other items:

- Overtime engineering and inspection.
- Special inspections, including vacuum excavation pot-holing.
- Survey materials, stakes, etc.
- Additional required services.

Any changes in the Consultant's or Subconsultant's staff or fee structure will be presented in writing to the City for approval prior to initiating any changes or performing any work.

Exhibit A



Price Proposal for Cheyenne Dry Creek Union Pacific Crossing City of Cheyenne, Wyoming | RFLP 1-25

Project Task and/or Subtask	Project Task or Subtask Description	GLM Design			ICON Engineering		Quandel Consultants		GEI	Labor Costs	Administrative	Total Cost
		GLM Design Group	GLM Design / Steve Pardue	GLM / Survey Will Morrow (BenchMark)	Jaclyn Michaelisen	Shuang Suang	Quandel Engineers	DJ Mitchell	Lance Heyer			
	Estimated Rates per Hour	\$168	\$175	\$105	\$210	\$165					\$75	
1	Kick-Off Meetings, Progress Meetings with City Staff, Preliminary Coordination with UPRR									\$6,170.00	\$900.00	\$7,070.00
A	Assume (1) Kick-Off Meeting and (6) Progress Meetings	7.00			7.00					\$2,646.00	8.00	\$3,246.00
B	Assume (2) Meetings with UPRR for Preliminary Railroad Coordination	5.00					\$1,484.00	\$1,200.00		\$3,524.00	4.00	\$3,824.00
Task 1: Feasibility Analysis												
2	Information Review and Topographic Survey									\$28,148.00	\$150.00	\$28,298.00
A	Data Collection, Field Surveys, geotechnical data, and Reconnaissance needed for H&H Modeling of Crossing Improvements	16.00		60.00	4.00				\$18,320.00	\$28,148.00	2.00	\$28,298.00
3	Hydraulic Model Development									\$73,464.00	\$750.00	\$74,214.00
A	Pare Down SubBasins in Adopted EPA Dry Creek SWMM Model for input into HEC-RAS Model and QA/QC for Model Stability	60.00			8.00	8.00				\$13,080.00	2.00	\$13,230.00
B	Alternative Crossing Improvements Model Development with and without Hydraulic Isolation of Historic UPRR 6' x 8' Box Arch Masonry Culvert	128.00			8.00	12.00				\$25,164.00	4.00	\$25,464.00
C	Use FWHY HY-8 Software to Calibrate Culvert Rating Curves	32.00				4.00				\$6,036.00	2.00	\$6,186.00
D	Calculate MPF Hydrology for Supplemental Breach Modeling with Calibrated HEC-RAS Model	88.00		16.00	48.00	16.00				\$29,184.00	2.00	\$29,334.00
Task 2: Benefit-Cost Analysis in Conjunction with Task 3												
4	Conceptual Alternatives Development with Cost Estimates and BCAs									\$29,316.00	\$1,050.00	\$30,366.00
A	Benefit-Cost Analyses of Conveyance Alternatives	40.00	28.00							\$11,620.00	4.00	\$11,920.00
B	EHP Evaluation, Coordination with city staff	24.00								\$4,032.00	8.00	\$4,632.00
C	Preparation of BCA Report	48.00	32.00							\$13,664.00	2.00	\$13,814.00
Task 3: Preliminary Design												
5	Conceptual Alternatives Development with Cost Estimates and BCAs									\$96,752.00	\$600.00	\$97,352.00
A	Design of Conveyance Alternatives / Estimate of Construction Cost	88.00			12.00	20.00	\$12,000.00			\$32,604.00	2.00	\$32,754.00
B	Alternative 1: Engineering Design and 30% Level Plans	88.00			8.00	20.00	\$12,000.00			\$31,764.00	2.00	\$31,914.00
C	Alternative 2: Engineering Design and 30% Level Plans	88.00			8.00	8.00	\$7,500.00			\$25,284.00	2.00	\$25,434.00
D	Quality Control / Quality Assurance (QA/QC) Compliance with UPRR Standards and Requirements	4.00			2.00		\$2,500.00			\$3,592.00		\$3,592.00
E	Project Team Meetings / Coordination with Quandel Consultants and GEI Consultants	6.00					\$1,250.00		\$1,250.00	\$3,508.00	2.00	\$3,658.00
Preliminary Design Documents												
6	Final Report									\$6,776.00	\$300.00	\$7,076.00
A	Preparation of Final Report Deliverables	24.00								\$4,032.00	2.00	\$4,182.00
B	Engineers Opinion of Cost for Recommended Alternatives/BCA Analyses	8.00	8.00							\$2,744.00		\$2,744.00
C	Preparation of Grant Strategy and FEMA BRIC Application for Preferred Alternative									\$0.00	2.00	\$150.00
Expenses												
7	Travel, Meals, Lodging, and Mileage						\$2,448.80			\$2,448.80	2.00	\$2,598.80
		754.00	68.00	76.00	105.00	88.00	\$39,182.80	\$1,200.00	\$19,570.00	\$243,074.80	52.00	\$246,974.80
	#####	\$11,900.00	\$7,980.00	\$22,050.00	\$14,520.00	\$39,182.80	\$1,200.00	\$19,570.00		\$3,900.00		\$246,974.80